Probabilistic Data Exchange

Information Integration 25/26

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Probabilistic Space over U x W

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Probability Space:

$\tilde{P} = (\Omega(\tilde{P}), P_{\tilde{P}})$

Sample Space:

$\Omega(\tilde{P}) = U \times W$

Left-marginal (subspace):

$\tilde{U}: \Omega(\tilde{U}) = U, \forall u \in U \ P_{\tilde{U}}(u) = \Sigma_{w \in W} P_{\tilde{P}}(u,w)$

Right-marginal (subspace):

$\tilde{W}: \Omega(\tilde{W}) = W, \forall w \in W \ P_{\tilde{W}}(w) = \Sigma_{u \in U} P_{\tilde{P}}(u,w)$

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Mapping of P-Databases

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Probabilistic database (P-instance): Probability Space over $Inst^c(R)$

$Inst^c(R)$: All ground (complete) database instances defined over schema R

Given a Mapping **M = (S, T, Σ)** [source, target schema, set of dependencies formulated as logical assertions over S and T]

Source p-instance $\tilde{I}$: ground p-instance over S.
Target p-instance $\tilde{J}$: p-instance over T.

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Probabilistic Match

Probabilistic Match

$\tilde{\mathbf{U}}$, $\tilde{\mathbf{W}}$: probability spaces
R $\subseteq \Omega(\tilde{U})$ x $\Omega(\tilde{W})$: binary relation

R-match of $\tilde{U}$ in $\tilde{W}$: p-space $\tilde{\mathbf{P}}$ over $\Omega(\tilde{U})$ x $\Omega(\tilde{W})$:

1. Left marginal of $\tilde{\mathbf{P}}$ is $\tilde{\mathbf{U}}$
2. Right marginal of $\tilde{\mathbf{P}}$ is $\tilde{\mathbf{W}}$
3. Pr(P $\in$ **R**) = 1

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(Universal) P-Solution

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M: mapping $\tilde{\mathbf{I}}$: source p-instance $\tilde{\mathbf{J}}$: target p-instance

$\mathbf{Sol_M}$ = {(I, J) | J solution for M, I}
$\mathbf{USol_M}$ = {(I, J) | J universal solution for M, I}

$\tilde{J}$ is **p-solution** for M, $\tilde{I} \triangleq \exists$ Sol_M-match of $\tilde{I}$ in $\tilde{J}$
$\tilde{J}$ is **universal p-solution** for M, $\tilde{I} \triangleq \exists$ $USol_M$-match of $\tilde{I}$ in $\tilde{J}$

Can we correlate universal p-solutions to p-solutions by means of the homomorphism?

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P-Instance Homomorphism Match

P-Instance Homomorphism Match

$\mathbf{HOM_T}$: $\{(J_1, J_2) \mid (J_1, J_2) \in (\mathrm{Inst}(T))^2;\ \exists\, h: \mathbf{J_1} \mathbf{\text{->}} \mathbf{J_2}\}$

Given $\mathbf{\tilde{J}_1}$, $\mathbf{\tilde{J}_2}$ p-instances over schema **T**

P-Instance Homomorphism Match of J̃1 in J̃2:

$\mathbf{\tilde{J}_1 \text{->}^M \tilde{J}_2} \triangleq \exists$ $\mathrm{HOM_T}$-match of $\tilde{J}_1$ in $\tilde{J}_2$

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Stochastic Extensions

Stochastic Extensions

J′ is at most specific as J

$\mathbf{J \leq_{SP} J'} \triangleq \exists h\ J \to J'$

J′ is at most generic as J

$\mathbf{J \leq_{GE} J'} \triangleq \exists h\ J' \to J$

$\tilde{J}_1$ is at most specific as $\tilde{J}_2$

$\mathbf{\tilde{J}_1 \to^{\leq SP} \tilde{J}_2} \triangleq Pr_{\tilde{J}1}(J_1 \to J) \geq Pr_{\tilde{J}2}(J_2 \to J)\ \forall J \in Inst(R)$

$\tilde{J}_1$ is at least generic as $\tilde{J}_2$

$\mathbf{\tilde{J}_1 \to^{\geq GE} \tilde{J}_2} \triangleq Pr_{\tilde{J}2}(J \to J_2) \geq Pr_{\tilde{J}1}(J \to J_1)\ \forall J \in Inst(R)$

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Universal Solution Theorem

Universal Solution Theorem

M: mapping **$\tilde{I}$**: source p-instance **$\tilde{J}$**: target p-solution

The following are equivalent:

(1) $\tilde{J}$ is a universal p-solution for M, $\tilde{I}$

(2) $\tilde{J} \to^{M} \tilde{J}'$ $\forall J'$ p-solution

(3) $\tilde{J} \to^{\leq SP} \tilde{J}'$ $\forall J'$ p-solution

(4) $\tilde{J} \to^{\geq GE} \tilde{J}'$ $\forall J'$ p-solution

(5) every Sol_M-match of $\tilde{I}$ in $\tilde{J}$ is a $USol_M$-match of $\tilde{I}$ in $\tilde{J}$

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Querying P-Databases

Querying P-Databases

Deterministic case:

$\text{cert}(Q, I, \Sigma) = \{a \in \text{Const}^k \mid a \in Q(J), \forall \text{ solutions } J\}$

Probabilistic case:

$\text{conf}_{Q, \tilde{I}, \Sigma}(a) = \inf_{\text{p-solution } \tilde{J}} \Pr_{\tilde{J}}(a \in Q(J))$

Theorem [applicable if ∃ p-solution $\tilde{J}$ for $\tilde{I}$, Σ]:

$\text{conf}_{Q, \tilde{I}, \Sigma}(a) = \Pr_{\tilde{I}}(a \in \text{cert}(Q, I, \Sigma))$

Querying P-Databases

Given a schema Mapping **M = (S, T, Σ)**

Ĩ: Source p-instance

Universal p-solutions can be used to compute probabilistic queries:

J̃: universal p-solution, **Q**: UCQ over T

$\mathrm{conf}_{Q,\tilde{I},\Sigma}(a) = \Pr_{\tilde{J}}(a \in Q(J))$

If a solution can be used to answer confidence queries for all boolean conjunctive queries that solution is universal:

J̃ p-solution: $\mathrm{conf}_{Q,\tilde{I},\Sigma}(a) = \Pr_{\tilde{J}}(Q(J))$, ∀ boolean CQs Q =>

J̃ is a universal p-solution

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Annotated instances

Annotated instances

EVar : $\{e_1, \ldots, e_n\}$ probabilistically independent boolean (Bernoulli) random variables.

DNF formula: $\phi_1 \lor \ldots \lor \phi_m$, where ϕ_i is conj (or neg.) of event variables.

DNF instance $I^{\alpha, p}$ $\triangleq$ p-instance over R

1) α: fact of R **f** -> α(f) dnf formula over Evar
2) p: EVar -> [0,1]

Th) Every finite p-instance can be represented by some DNF instance $I^{\alpha, p}$

Annotated instances

Tuple-independent instance $I^{\alpha, p}$ $\triangleq$ DNF instance: $\forall f \in I$ (facts) the condition $\alpha(f)$ is a distinct atomic event variable e_f.

Possible elements of **Σ**: [ψ, ϕ conjunction of atomic formulas over s, t]

ST-TGD $\triangleq \forall x\ (\phi_s(x) \rightarrow \exists y\ \psi_t(x,y))$

T-TGD $\triangleq \forall x\ (\phi_t(x) \rightarrow \exists y\ \psi_t(x,y))$

T-EGD $\triangleq \forall x\ (\phi_t(x) \rightarrow (x_1 = x_2))$

Full ST-TGD $\triangleq \forall x\ (\phi_s(x) \rightarrow \psi_t(x))$

Full T-TGD $\triangleq \forall x\ (\phi_t(x) \rightarrow \psi_t(x))$

Standard mapping: contains only st-tgds and t-egds + weakly acyclic set of t-tgds

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(some) Complexity Results

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Existence of (universal) p-solution: given a standard mapping a universal p-solution exists ⇔ a p-solution exists.
Deciding whether a p-solution exists co-NP, polynomial time: in case of tuple independent instance.

Materializing universal p-solution: assuming RP ≠ NP. ∃ schema mapping with only full st-tgd(s) and full t-tgd(s): ∄ polynomial randomized time algorithm that construct a universal p-solution as DNF instance $J^{\beta, q}$, given a tuple independent instance $I^{\alpha, p}$.

(some) Complexity Results

Query answering: Q is UCQ. Σ with only st-tgd(s). Given a source a tuple independent instance $I^{\alpha, p}$ and tuple a computing $conf_Q(a)$ is $FP^{\#P}$-complete [functional polynomial with an oracle solving countable problems]

Query answering: assuming RP ≠ NP. Q non trivial UCQ over T. ∃ schema mapping (S, T', Σ), T' ⊆ T. Σ contains only full st-tgd(s) and t-tgd(s): ∀θ there is no polynomial-time randomized θ-approximation algorithm to compute $conf_Q(a)$ over a source a for tuple independent instance $I^{\alpha, p}$.

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Probabilistic Mapping

Probabilistic Mapping

Dep_{ST}: countably infinite set of all formulas over <S, T>
$\Sigma \subseteq Dep_{ST}$: mapping [so far] is a subset of all those formulas
Dep^*_{ST}: countably set of all subset **Σ** of **Dep_{ST}**

Probabilistic Mapping: replace **Σ** with **$\tilde{\Sigma}$**, p-space over **Dep^*_{ST}**

p-problem: $\tilde{P}$ p-space over $Dep^*_{ST} \times Inst^C(S)$

Probabilistic Mapping

p-solution: $\tilde{J}$ for p-problem $\tilde{P}$ is $dSOL_{ST}$-match from $\tilde{P}$ to $\tilde{J}$

Binary relation between pairs (Σ, I) and instances J: $I \in Inst^{C}(S)$, $J \in Inst(T)$, $\Sigma \in Dep^{*}_{ST}$, $\langle I,J \rangle \vDash \Sigma$

universal p-solution: $\tilde{J}$ for p-problem $\tilde{P}$ is $dUSOL_{ST}$-match from $\tilde{P}$ to $\tilde{J}$

Binary relation between pairs (Σ, I) and instances J: J is a universal solution for I with respect to Σ

Previous complexity results generalize to p-problem!

Q&A

Thank You!

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